T Pranav Kutralingam

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Experience

Pretorin Mar 2026 – Sep 2026
Software Engineer (Product) *Tempe, AZ*

- Built a production agentic RAG chatbot from zero to one over **1,000+ documents per customer** for government compliance teams, and raised retrieval precision from **71% to 93%**.
- Built guardrails as code that stop the LLM from writing compliance control text, plus an LLM-judged eval suite in CI (**85% agreement** with human labels) that blocked regressions.
- Built ingestion for PDFs, images and text handling up to **28 upstream document updates a day**. GPT-4o vision turns image evidence into validated JSON.
- Fixed an **89% upload failure rate** at 10 concurrent users, found with a Locust load test I wrote, and added a CI lint rule that blocked the bug class.

DaRL Lab, Arizona State University May 2024 – Dec 2025
Graduate Research Assistant *Tempe, AZ*

- Cut LLM eval sweeps from **12 hours to under 1 hour** with a concurrent Python scheduler across 8 model backends, including vLLM inference on A100 GPUs.
- Built the LLM eval harness for **ACL 2026** and the multi-GPU PyTorch RL training for **ICLR 2026**.
- Made 2,000-job distributed sweeps on a 10-GPU Slurm cluster fault-tolerant, so experiments that took **4–5 days finished overnight**.

Amadeus Jul 2021 – Dec 2023
Software Engineer *Bengaluru, India*

- Built a gradient-boosting ML pipeline that flags likely false third-party Agency Debit Memo claims for dispute, saving **$500K a year**.
- Shipped the IndiGo integration into Amadeus's travel-agent booking platform with Spring Boot REST and payment APIs and idempotent checkout. India market share grew **18%**.
- Replaced manual triage of **15K+ monthly test failures** with a scikit-learn classifier at 97% precision, so releases moved from every 2–3 days to **daily**.
- Designed an event-driven Java service that syncs itineraries for a consumer travel app at **5 ms p99 latency** on peak days of 100K+ requests.
- Led 5 engineers on internal tooling used by **12 teams**: planned the projects, reviewed code and mentored.

Projects & Publications

Lost in Execution: On the Multilingual Robustness of Tool Calling in LLMs ACL 2026

- Benchmarks how accurately **13 LLMs** call tools across 4 languages. Python harness serves open models on vLLM, parses each tool call and scores it against ground truth.

Latent Adaptation of Foundation Policies for Sim-to-Real Transfer ICLR 2026

- Studies adapting pretrained RL policies for sim-to-real transfer. Training ran **9 policies** concurrently with checkpointing and auto-resume.

HEAL AI: Medical Insurance Assistant | *Python, FastAPI, React, TypeScript, Gemini* 2025

- Answers questions on uploaded medical bills and insurance policies and flags coverage gaps and overcharges. A React client calls a FastAPI RAG backend that grounds Gemini in the user's files.

Technical Skills

AI & ML: RAG, AI Agents, Tool Calling, MCP, LLM Evals, Guardrails, vLLM, LangGraph, PyTorch, QLoRA
Languages & Frameworks: Python, Java, TypeScript, SQL, FastAPI, Spring Boot, React, Pytest, Playwright
Infrastructure: AWS, Azure, Docker, Kubernetes, Terraform, Slurm, GitHub Actions, PostgreSQL, Splunk
AI-Assisted Development: GitHub Copilot, Claude Code, Cursor

Education

Arizona State University, M.S. Computer Science, GPA 3.97/4.0 Jan 2024 – Dec 2025
SRM Institute of Science and Technology, B.Tech Computer Science May 2017 – May 2021